

The following additional libraries are needed to run this notebook. Note that running on Colab is experimental, please report a Github issue if you have any problem.

```
!pip install d2l==1.0.3
```

The Transformer Architecture

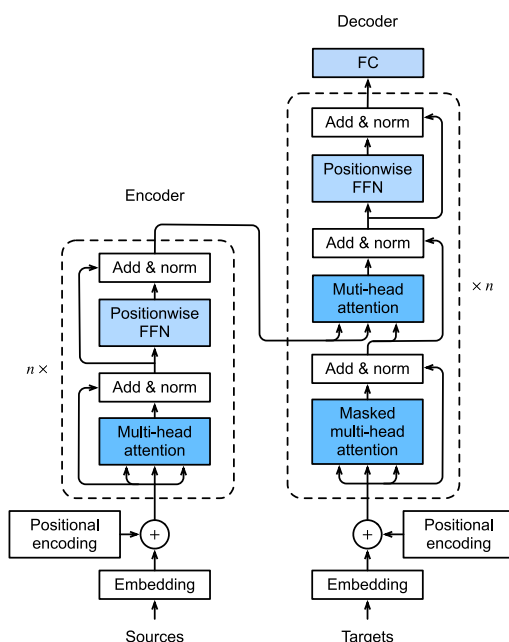
:label: sec_transformer

We have compared CNNs, RNNs, and self-attention in :numref: subsec_cnn-rnn-self-attention. Notably, self-attention enjoys both parallel computation and the shortest maximum path length. Therefore, it is appealing to design deep architectures by using self-attention. Unlike earlier self-attention models that still rely on RNNs for input representations :cite: Cheng.Dong.Lapata.2016, Lin.Feng.Santos.ea.2017, Paulus.Xiong.Socher.2017, the Transformer model is solely based on attention mechanisms without any convolutional or recurrent layer :cite: Vaswani.Shazeer.Parmar.ea.2017. Though originally proposed for sequence-to-sequence learning on text data, Transformers have been pervasive in a wide range of modern deep learning applications, such as in areas to do with language, vision, speech, and reinforcement learning.

```
import math
import pandas as pd
import torch
from torch import nn
from d2l import torch as d2l
```

Model

As an instance of the encoder--decoder architecture, the overall architecture of the Transformer is presented in :numref: fig_transformer. As we can see, the Transformer is composed of an encoder and a decoder. In contrast to Bahdanau attention for sequence-to-sequence learning in :numref: fig_s2s_attention_details, the input (source) and output (target) sequence embeddings are added with positional encoding before being fed into the encoder and the decoder that stack modules based on self-attention.



:width: 320px :label: fig_transformer

Now we provide an overview of the Transformer architecture in :numref: fig_transformer. At a high level, the Transformer encoder is a stack of multiple identical layers, where each layer has two sublayers (either is denoted as sublayer). The first is a multi-head self-attention pooling and the second is a positionwise feed-forward network. Specifically, in the encoder self-attention, queries, keys, and values are all from the outputs of the previous encoder layer. Inspired by the ResNet design of :numref: sec_resnet, a residual connection is employed around both sublayers. In the Transformer, for any input $\mathbf{x} \in \mathbb{R}^d$ at any position of the sequence, we require that $\text{sublayer}(\mathbf{x}) \in \mathbb{R}^d$ so that the residual connection $\mathbf{x} + \text{sublayer}(\mathbf{x}) \in \mathbb{R}^d$ is feasible. This addition from the residual connection is immediately followed by layer normalization :cite: Ba.Kiros.Hinton.2016. As a result, the Transformer encoder outputs a d -dimensional vector representation for each position of the input sequence.

The Transformer decoder is also a stack of multiple identical layers with residual connections and layer normalizations. As well as the two sublayers described in the encoder, the decoder inserts a third sublayer, known as the encoder--decoder attention, between these two. In the encoder--decoder attention, queries are from the outputs of the decoder's self-attention sublayer, and the keys and values are from the Transformer encoder outputs. In the decoder self-attention, queries, keys, and values are all from the outputs of the previous decoder layer. However, each position in the decoder is allowed only to attend to all positions in the decoder up to that position. This *masked* attention preserves the autoregressive property, ensuring that the prediction only depends on those output tokens that have been generated.

We have already described and implemented multi-head attention based on scaled dot products in :numref:sec_multihead-attention and positional encoding in :numref:subsec_positional-encoding. In the following, we will implement the rest of the Transformer model.

[Positionwise Feed-Forward Networks]

:label:subsec_positionwise-ffn

The positionwise feed-forward network transforms the representation at all the sequence positions using the same MLP. This is why we call it *positionwise*. In the implementation below, the input X with shape (batch size, number of time steps or sequence length in tokens, number of hidden units or feature dimension) will be transformed by a two-layer MLP into an output tensor of shape (batch size, number of time steps, ffn_num_outputs).

```
class PositionWiseFFN(nn.Module): #@save
    """The positionwise feed-forward network."""
    def __init__(self, ffn_num_hiddens, ffn_num_outputs):
        super().__init__()
        self.dense1 = nn.Linear(ffn_num_hiddens)
        self.relu = nn.ReLU()
        self.dense2 = nn.Linear(ffn_num_hiddens)

    def forward(self, X):
        return self.dense2(self.relu(self.dense1(X)))
```

"@save" is not an allowed annotation - allowed values include [@param, @title, @markdown].

[Hide code](#)

The following example shows that [the innermost dimension of a tensor changes] to the number of outputs in the positionwise feed-forward network. Since the same MLP transforms at all the positions, when the inputs at all these positions are the same, their outputs are also identical.

```
ffn = PositionWiseFFN(4, 8)
ffn.eval()
ffn(torch.ones((2, 3, 4)))[0]
```

```
tensor([[ 0.6300,  0.7739,  0.0278,  0.2508, -0.0519,  0.4881, -0.4105,  0.5163],
        [ 0.6300,  0.7739,  0.0278,  0.2508, -0.0519,  0.4881, -0.4105,  0.5163],
        [ 0.6300,  0.7739,  0.0278,  0.2508, -0.0519,  0.4881, -0.4105,  0.5163]],
        grad_fn=<SelectBackward0>)
```

Residual Connection and Layer Normalization

Now let's focus on the "add & norm" component in :numref:fig_transformer. As we described at the beginning of this section, this is a residual connection immediately followed by layer normalization. Both are key to effective deep architectures.

In :numref:sec_batch_norm, we explained how batch normalization recenters and rescales across the examples within a minibatch. As discussed in :numref:subsec_layer-normalization-in-bn, layer normalization is the same as batch normalization except that the former normalizes across the feature dimension, thus enjoying benefits of scale independence and batch size independence. Despite its pervasive applications in computer vision, batch normalization is usually empirically less effective than layer normalization in natural language processing tasks, where the inputs are often variable-length sequences.

The following code snippet [compares the normalization across different dimensions by layer normalization and batch normalization].

```
ln = nn.LayerNorm(2)
bn = nn.BatchNorm1d()
X = torch.tensor([[1, 2], [2, 3]], dtype=torch.float32)
# Compute mean and variance from X in the training mode
print('layer norm:', ln(X), '\nbatch norm:', bn(X))
```

```
layer norm: tensor([[ -1.0000,  1.0000],
                   [-1.0000,  1.0000]], grad_fn=<NativeLayerNormBackward0>)
batch norm: tensor([[ -1.0000, -1.0000],
                   [ 1.0000,  1.0000]], grad_fn=<NativeBatchNormBackward0>)
```

Now we can implement the `AddNorm` class [using a residual connection followed by layer normalization]. Dropout is also applied for regularization.

```
class AddNorm(nn.Module): #@save
    """The residual connection followed by layer normalization"""
    def __init__(self, norm_shape, dropout):
        super().__init__()
        self.dropout = nn.Dropout(dropout)
        self.ln = nn.LayerNorm(norm_shape)

    def forward(self, X, Y):
        return self.ln(self.dropout(Y) + X)
```

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The residual connection requires that the two inputs are of the same shape so that [the output tensor also has the same shape after the addition operation].

```
add_norm = AddNorm(4, 0.5)
shape = (2, 3, 4)
d2l.check_shape(add_norm(torch.ones(shape), torch.ones(shape)), shape)
```

Encoder

:label: subsec_transformer-encoder

With all the essential components to assemble the Transformer encoder, let's start by implementing [a single layer within the encoder]. The following `TransformerEncoderBlock` class contains two sublayers: multi-head self-attention and positionwise feed-forward networks, where a residual connection followed by layer normalization is employed around both sublayers.

```
class TransformerEncoderBlock(nn.Module): #@save
    """The Transformer encoder block."""
    def __init__(self, num_hiddens, ffn_num_hiddens, num_heads,
                 use_bias=False):
        super().__init__()
        self.attention = d2l.MultiHeadAttention(num_hiddens, num_hiddens, num_heads, dropout, use_bias)
        self.addnorm1 = AddNorm(num_hiddens, dropout)
        self.ffn = PositionWiseFFN(ffn_num_hiddens, num_hiddens)
        self.addnorm2 = AddNorm(num_hiddens, dropout)

    def forward(self, X, valid_lens):
        Y = self.addnorm1(X, self.attention(X, X, X, valid_lens))
        return self.addnorm2(Y, self.ffn(Y))
```

"@save" is not an allowed annotation - allowed values include [`@param`, `@title`, `@markdown`].

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As we can see, [no layer in the Transformer encoder changes the shape of its input.]

```
X = torch.ones((2, 100, 24))
valid_lens = torch.tensor([3, 2])
encoder_blk = TransformerEncoderBlock(24, 48, 8, 0.5)
encoder_blk.eval()
d2l.check_shape(encoder_blk(X, valid_lens), X.shape)
```

In the following [Transformer encoder] implementation, we stack `num_blks` instances of the above `TransformerEncoderBlock` classes. Since we use the fixed positional encoding whose values are always between -1 and 1 , we multiply values of the learnable input embeddings by the square root of the embedding dimension to rescale before summing up the input embedding and the positional encoding.

```
class TransformerEncoder(d2l.Encoder): #@save
    """The Transformer encoder."""
    def __init__(self, vocab_size, num_hiddens, ffn_num_hiddens, num_heads, num_blks, dropout, use_bias=False):
        super().__init__()
        self.num_hiddens = num_hiddens
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = d2l.PositionalEncoding(num_hiddens)
        self.blks = nn.Sequential()
        for i in range(num_blks):
            self.blks.add_module("block"+str(i), TransformerEncoderBlock(num_hiddens, ffn_num_hiddens, num_heads, dropout, use_bias))

    def forward(self, X, valid_lens):
        # Since positional encoding values are between -1
```

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```

# values are multiplied by the square root of the
# to rescale before they are summed up
X = self.pos_encoding(self.embedding(X) * math.sqrt(
self.attention_weights = [None] * len(self.blks)
for i, blk in enumerate(self.blks):
    X = blk(X, valid_lens)
    self.attention_weights[
        i] = blk.attention.attention.attention_weights
return X

```

Below we specify hyperparameters to [create a two-layer Transformer encoder]. The shape of the Transformer encoder output is (batch size, number of time steps, `num_hiddens`).

```

encoder = TransformerEncoder(200, 24, 48, 8, 2, 0.5)
d2l.check_shape(encoder(torch.ones((2, 100), dtype=torch.long), valid_lens),
(2, 100, 24))

```

Decoder

As shown in :numref:fig_transformer, [the Transformer decoder is composed of multiple identical layers]. Each layer is implemented in the following `TransformerDecoderBlock` class, which contains three sublayers: decoder self-attention, encoder-decoder attention, and positionwise feed-forward networks. These sublayers employ a residual connection around them followed by layer normalization.

As we described earlier in this section, in the masked multi-head decoder self-attention (the first sublayer), queries, keys, and values all come from the outputs of the previous decoder layer. When training sequence-to-sequence models, tokens at all the positions (time steps) of the output sequence are known. However, during prediction the output sequence is generated token by token; thus, at any decoder time step only the generated tokens can be used in the decoder self-attention. To preserve autoregression in the decoder, its masked self-attention specifies `dec_valid_lens` so that any query only attends to all positions in the decoder up to the query position.

```

class TransformerDecoderBlock(nn.Module):
    # The i-th block in the Transformer decoder
    def __init__(self, num_hiddens, ffn_num_hiddens, num_heads, dropout, i):
        super().__init__()
        self.i = i
        self.attention1 = d2l.MultiHeadAttention(num_hiddens, num_heads,
                                                dropout)
        self.addnorm1 = AddNorm(num_hiddens, dropout)
        self.attention2 = d2l.MultiHeadAttention(num_hiddens, num_heads,
                                                dropout)
        self.addnorm2 = AddNorm(num_hiddens, dropout)
        self.ffn = PositionWiseFFN(ffn_num_hiddens, num_hiddens)
        self.addnorm3 = AddNorm(num_hiddens, dropout)

    def forward(self, X, state):
        enc_outputs, enc_valid_lens = state[0], state[1]
        # During training, all the tokens of any output sequence are processed
        # at the same time, so state[2][self.i] is None as initialized. When
        # decoding any output sequence token by token during prediction,
        # state[2][self.i] contains representations of the decoded output at
        # the i-th block up to the current time step
        if state[2][self.i] is None:
            key_values = X
        else:
            key_values = torch.cat((state[2][self.i], X), dim=1)
        state[2][self.i] = key_values
        if self.training:
            batch_size, num_steps, _ = X.shape
            # Shape of dec_valid_lens: (batch_size, num_steps), where every
            # row is [1, 2, ..., num_steps]
            dec_valid_lens = torch.arange(
                1, num_steps + 1, device=X.device).repeat(batch_size, 1)
        else:
            dec_valid_lens = None
        # Self-attention
        X2 = self.attention1(X, key_values, key_values, dec_valid_lens)
        Y = self.addnorm1(X, X2)
        # Encoder-decoder attention. Shape of enc_outputs:
        # (batch_size, num_steps, num_hiddens)
        Y2 = self.attention2(Y, enc_outputs, enc_outputs, enc_valid_lens)
        Z = self.addnorm2(Y, Y2)
        return self.addnorm3(Z, self.ffn(Z)), state

```

To facilitate scaled dot product operations in the encoder--decoder attention and addition operations in the residual connections, **[the feature dimension (`num_hiddens`) of the decoder is the same as that of the encoder.]**

```
decoder_blk = TransformerDecoderBlock(24, 48, 8, 0.5, 0)
X = torch.ones((2, 100, 24))
state = [encoder_blk(X, valid_lens), valid_lens, [None]]
d2l.check_shape(decoder_blk(X, state)[0], X.shape)
```

Now we **[construct the entire Transformer decoder]** composed of `num_blks` instances of `TransformerDecoderBlock`. In the end, a fully connected layer computes the prediction for all the `vocab_size` possible output tokens. Both of the decoder self-attention weights and the encoder--decoder attention weights are stored for later visualization.

```
class TransformerDecoder(d2l.AttentionDecoder):
    def __init__(self, vocab_size, num_hiddens, ffn_num_hiddens, num_heads,
                  num_blks, dropout):
        super().__init__()
        self.num_hiddens = num_hiddens
        self.num_blks = num_blks
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = d2l.PositionalEncoding(num_hiddens, dropout)
        self.blks = nn.Sequential()
        for i in range(num_blks):
            self.blks.add_module("block"+str(i), TransformerDecoderBlock(
                num_hiddens, ffn_num_hiddens, num_heads, dropout, i))
        self.dense = nn.LazyLinear(vocab_size)

    def init_state(self, enc_outputs, enc_valid_lens):
        return [enc_outputs, enc_valid_lens, [None] * self.num_blks]

    def forward(self, X, state):
        X = self.pos_encoding(self.embedding(X) * math.sqrt(self.num_hiddens))
        self._attention_weights = [[None] * len(self.blks) for _ in range(2)]
        for i, blk in enumerate(self.blks):
            X, state = blk(X, state)
            # Decoder self-attention weights
            self._attention_weights[0][i] = blk.attention1.attention.attention_weights
            # Encoder-decoder attention weights
            self._attention_weights[1][i] = blk.attention2.attention.attention_weights
        return self.dense(X), state

    @property
    def attention_weights(self):
        return self._attention_weights
```

✓ [Training]

Let's instantiate an encoder--decoder model by following the Transformer architecture. Here we specify that both the Transformer encoder and the Transformer decoder have two layers using 4-head attention. As in :numref:sec_seq2seq_training, we train the Transformer model for sequence-to-sequence learning on the English--French machine translation dataset.

```
data = d2l.MTFraEng(batch_size=128)
num_hiddens, num_blks, dropout = 256, 2, 0.2
ffn_num_hiddens, num_heads = 64, 4
encoder = TransformerEncoder(
    len(data.src_vocab), num_hiddens, ffn_num_hiddens, num_heads,
    num_blks, dropout)
decoder = TransformerDecoder(
    len(data.tgt_vocab), num_hiddens, ffn_num_hiddens, num_heads,
    num_blks, dropout)
model = d2l.Seq2Seq(encoder, decoder, tgt_pad=data.tgt_vocab['<pad>'],
                    lr=0.001)
trainer = d2l.Trainer(max_epochs=30, gradient_clip_val=1, num_gpus=1)
trainer.fit(model, data)
```

After training, we use the Transformer model to **[translate a few English sentences]** into French and compute their BLEU scores.

```
engs = ['go .', 'i lost .', 'he\'s calm .', 'i\'m home .']
fras = ['va !', 'j\'ai perdu .', 'il est calme .', 'je suis chez moi .']
preds, _ = model.predict_step(
```

```

data.build(engs, frs), d2l.try_gpu(), data.num_steps)
for en, fr, p in zip(engs, frs, preds):
    translation = []
    for token in data.tgt_vocab.to_tokens(p):
        if token == '<eos>':
            break
        translation.append(token)
    print(f'{en} => {translation}, bleu, '
          f'{d2l.bleu(" ".join(translation), fr, k=2):.3f}')

go . => ['va', '!'], bleu,1.000
i lost . => ['je', 'perdu', '.'], bleu,0.687
he's calm . => ['il', 'est', 'mouillé', '.'], bleu,0.658
i'm home . => ['je', 'suis', 'chez', 'moi', '.'], bleu,1.000

```

Let's [visualize the Transformer attention weights] when translating the final English sentence into French. The shape of the encoder self-attention weights is (number of encoder layers, number of attention heads, `num_steps` or number of queries, `num_steps` or number of key-value pairs).

```

_, dec_attention_weights = model.predict_step(
    data.build([engs[-1]], [fras[-1]]), d2l.try_gpu(), data.num_steps, True)
enc_attention_weights = torch.cat(model.encoder.attention_weights, 0)
shape = (num_blks, num_heads, -1, data.num_steps)
enc_attention_weights = enc_attention_weights.reshape(shape)
d2l.check_shape(enc_attention_weights,
                (num_blks, num_heads, data.num_steps, data.num_steps))

```

In the encoder self-attention, both queries and keys come from the same input sequence. Since padding tokens do not carry meaning, with specified valid length of the input sequence no query attends to positions of padding tokens. In the following, two layers of multi-head attention weights are presented row by row. Each head independently attends based on a separate representation subspace of queries, keys, and values.

```

d2l.show_heatmaps(
    enc_attention_weights.cpu(), xlabel='Key positions',
    ylabel='Query positions', titles=['Head %d' % i for i in range(1, 5)],
    figsize=(7, 3.5))

```

[To visualize the decoder self-attention weights and the encoder--decoder attention weights, we need more data manipulations.] For example, we fill the masked attention weights with zero. Note that the decoder self-attention weights and the encoder--decoder attention weights both have the same queries: the beginning-of-sequence token followed by the output tokens and possibly end-of-sequence tokens.

```

dec_attention_weights_2d = [head[0].tolist()
                           for step in dec_attention_weights
                           for attn in step for blk in attn for head in blk]
dec_attention_weights_filled = torch.tensor(
    pd.DataFrame(dec_attention_weights_2d).fillna(0.0).values)
shape = (-1, 2, num_blks, num_heads, data.num_steps)
dec_attention_weights = dec_attention_weights_filled.reshape(shape)
dec_self_attention_weights, dec_inter_attention_weights = \
    dec_attention_weights.permute(1, 2, 3, 0, 4)

```

```

d2l.check_shape(dec_self_attention_weights,
                (num_blks, num_heads, data.num_steps, data.num_steps))
d2l.check_shape(dec_inter_attention_weights,
                (num_blks, num_heads, data.num_steps, data.num_steps))

```

Because of the autoregressive property of the decoder self-attention, no query attends to key--value pairs after the query position.

```

d2l.show_heatmaps(
    dec_self_attention_weights[:, :, :, :],
    xlabel='Key positions', ylabel='Query positions',
    titles=['Head %d' % i for i in range(1, 5)], figsize=(7, 3.5))

```

Similar to the case in the encoder self-attention, via the specified valid length of the input sequence, [no query from the output sequence attends to those padding tokens from the input sequence.]

```
d2l.show_heatmaps(  
    dec_inter_attention_weights, xlabel='Key positions',  
    ylabel='Query positions', titles=['Head %d' % i for i in range(1, 5)],  
    figsize=(7, 3.5))
```

Although the Transformer architecture was originally proposed for sequence-to-sequence learning, as we will discover later in the book, either the Transformer encoder or the Transformer decoder is often individually used for different deep learning tasks.

✓ Summary

The Transformer is an instance of the encoder--decoder architecture, though either the encoder or the decoder can be used individually in practice. In the Transformer architecture, multi-head self-attention is used for representing the input sequence and the output sequence, though the decoder has to preserve the autoregressive property via a masked version. Both the residual connections and the layer normalization in the Transformer are important for training a very deep model. The positionwise feed-forward network in the Transformer model transforms the representation at all the sequence positions using the same MLP.

Exercises

1. Train a deeper Transformer in the experiments. How does it affect the training speed and the translation performance?
2. Is it a good idea to replace scaled dot product attention with additive attention in the Transformer? Why?
3. For language modeling, should we use the Transformer encoder, decoder, or both? How would you design this method?
4. What challenges can Transformers face if input sequences are very long? Why?
5. How would you improve the computational and memory efficiency of Transformers? Hint: you may refer to the survey paper by :citet:Tay.Dehghani.Bahri.ea.2020.

[Discussions](#)